Project 3: Hybrid RAG for BFSI Customer Risk, Retention & Next-Best-Action Intelligence

Project brief converted from the provided project sheet

Field	Details
Industry / Business Area	Banking & Financial Services; Customer Retention & Risk Modelling; Segmentation; Personalization; Customer Journey Analytics
Technology Theme	Hybrid RAG combining knowledge graph relationships, semantic search, structured customer data, rule checks, explainable recommendations, risk controls, and human approval
Project Description	Build a hybrid intelligence assistant for a bank or financial-services client. The system should help relationship managers or service teams understand customer risk, churn likelihood, complaint history, product fit, and next-best-action recommendations using a mix of structured data, interaction notes, policy documents, and behavioral signals.
Input Data / Assets	Synthetic banking customer profiles, transaction-event summaries, complaint logs, call-center notes, product catalog, retention offers, policy documents, regulatory guidance snippets
Evaluation Metrics	Correct risk-signal identification, recommendation precision, groundedness, compliance-safe response rate, false-positive/false-negative risk flags, bias checks, RM productivity improvement

High-Level Work Areas

#	Work Area
1	Create synthetic customer profiles, account/product holdings, complaint history, transaction events, service interactions, and policy documents.
2	Build a baseline rules-based risk and retention summary.
3	Add semantic retrieval over policy, complaint notes, and interaction summaries.
4	Add knowledge graph linking customer, products, complaints, journeys, risk signals, and recommended actions.
5	Build hybrid retrieval for customer context + policy grounding.
6	Generate next-best-action with rationale, evidence, risk warnings, and compliance constraints.
7	Add PII masking, access controls, and human approval workflow.
8	Evaluate recommendations for correctness, groundedness, compliance, and bias.

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Progressive Demo Script

Step	Demo Sequence / Talking Point
1	Start with a customer question: 'Why is this premium customer at risk of churn and what should the RM do next?'
2	Show a simple rules-based scorecard using balance drop, complaints, and product usage. Highlight that rules are explainable but miss context from notes and policies.
3	Add LLM summarization of customer notes only. Highlight that it summarizes well but may recommend actions not permitted by policy.
4	Add RAG over product and policy documents; ask: 'Which retention offers are allowed for this customer profile?' Show policy grounding.
5	Add structured customer context retrieval; ask: 'Summarize risk drivers across complaints, transactions, and engagement.' Show richer but still fragmented reasoning.
6	Add knowledge graph relationships: customer -> products -> complaints -> touchpoints -> risk signals -> offers. Ask for the evidence path behind churn risk.
7	Add hybrid retrieval combining structured filters, graph paths, and semantic policy retrieval. Ask for a final next-best-action with evidence and constraints.
8	Add compliance guardrails: PII masking, no unauthorized advice, mandatory escalation when risk is high. Demonstrate a blocked/flagged recommendation.
9	Add HITL approval: RM can accept, edit, or reject the recommendation, feeding evaluation data.
10	Final limitations to highlight: synthetic data may not reflect real bias, causality is not guaranteed, policy changes require freshness controls, and high-risk financial advice needs governance.
11	Future scope: real-time event streaming, calibrated churn model integration, offer uplift modeling, fairness testing, CRM integration, and audit-ready recommendation logs.